Given: Due:	Further Maths (Decision) HW 04	Name:

Chapter 1: Algorithms

You MUST refer to class notes and complete personal study before attempting this homework. Attempt EVERY part of EVERY question. Check your work for errors or omissions before handing in.

91% (A*)	76% (A)	66% (B)	56% (C)	50% (D)	40% (E)

Useful resources: *Useful resources*: Your notes from recent lessons & your text book

Questions - do these on lined paper

1)

42 21 15 16 35 10 31 11 27 39

(a) Use the first-fit bin packing algorithm to determine how the numbers listed above can be packed into bins of size 65

(3)

(b) The list of numbers is to be sorted into descending order. Use a quick sort to obtain the sorted list. You should show the result of each pass and identify your pivots clearly.

(4)

(c) Use the first-fit decreasing bin packing algorithm on your ordered list to pack the numbers into bins of size 65

(3)

The nine distinct numbers below are to be sorted into descending order

23 14 17 x 21 18 8 20 11

A bubble sort, starting at the left-hand end of the list, is to be used to obtain the sorted list. After the first complete pass, the list is

23 17 x 21 18 14 20 11 8

After the second complete pass, the list is

23 17 21 18 x 20 14 11 8

(d) Using this information, write down the smallest interval that must contain x. Give your answer as an inequality.

(3)

2)

59 45 18 55 47 11 63 17 15 42

(a) The list of numbers above is to be sorted into descending order. Perform a quick sort to obtain the sorted list. You should show the result of each pass and identify your pivots clearly.

(4)

The numbers in the list represent the lengths, in cm, of some pieces of copper wire. The copper wire is sold in one metre lengths.

(b) Use the first-fit decreasing bin packing algorithm to determine how these pieces could be cut from one metre lengths. (You should ignore wastage due to cutting.)

(3)

(c) Determine whether your solution to (b) is optimal. Give a reason for your answer.

(2)

3)

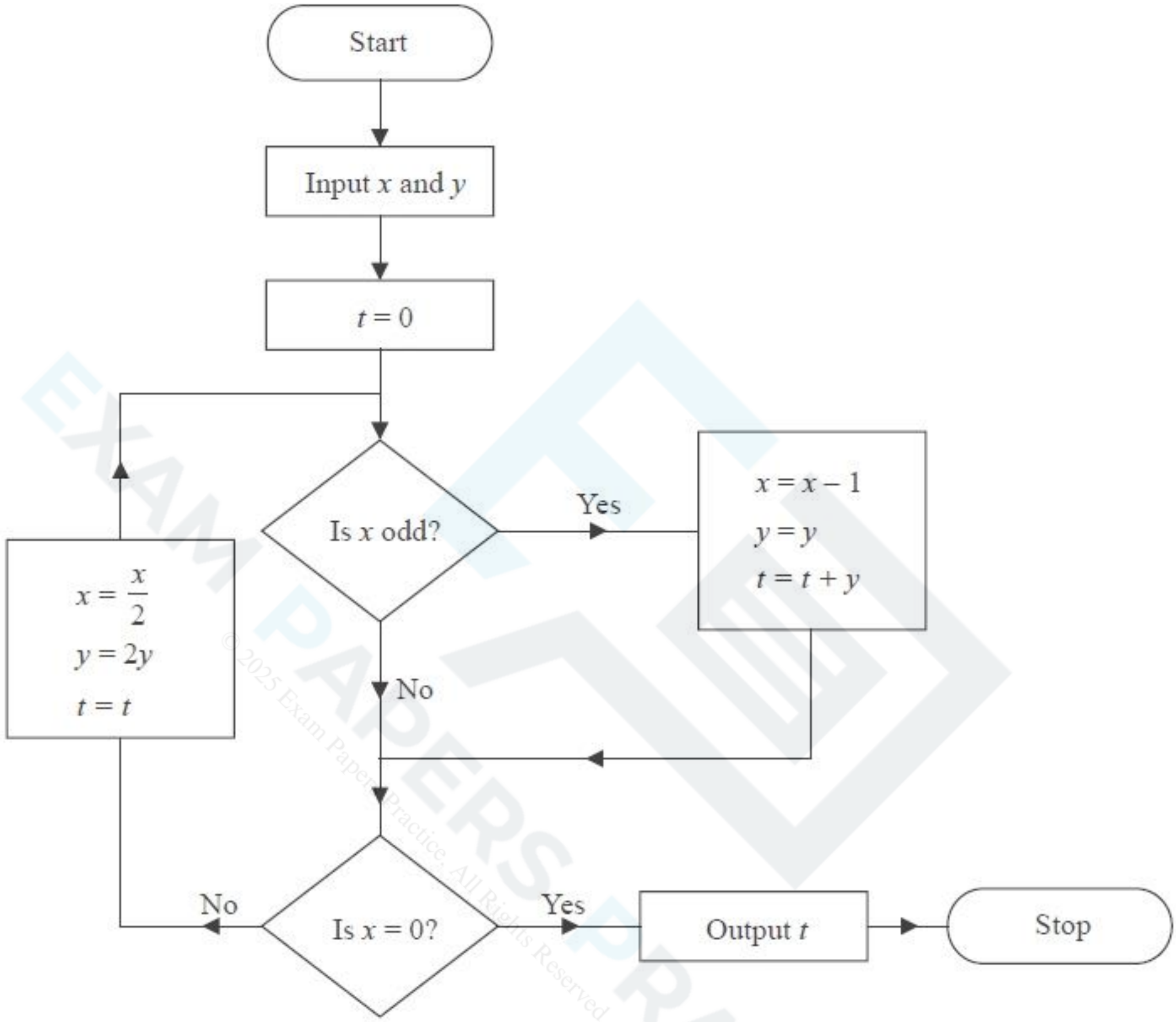

Figure 4

An algorithm is described by the flow chart shown in Figure 4.

Given that x = 27 and y = 5,

(a) complete the table in the answer book to show the results obtained at each step when the algorithm is applied. Give the final output.

(4)

The numbers 122 and $\frac{1}{2}$ are to be used as inputs for the algorithm described by the flow chart.

(b) (i) State, giving a reason, which number should be input as x.

(ii) State the output.

(3)

4)

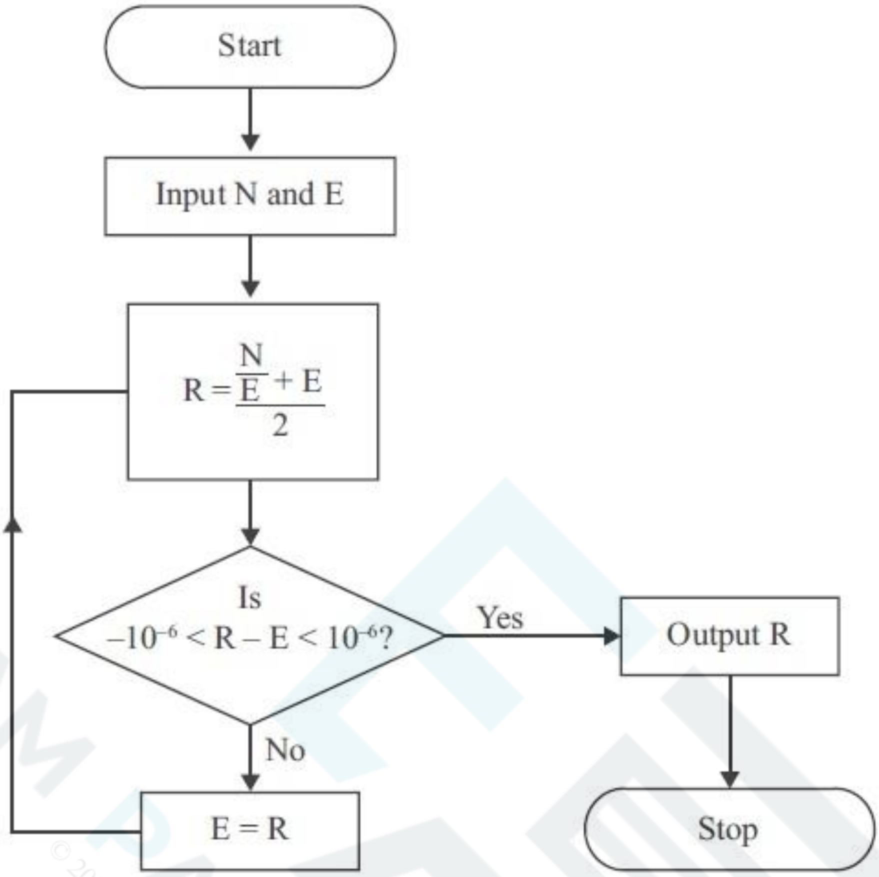

Figure 1

Hero's algorithm for finding a square root is described by the flow chart shown in Figure 1.

Given that N = 72 and E = 8,

(a) use the flow chart to complete the table in the answer book, working to at least seven decimal places when necessary. Give the final output correct to seven decimal places.

(4)

The flow chart is used with N = 72 and E = −8,

(b) describe how this would affect the output.

(1)

(c) State the value of E which cannot be used when using this flow chart.

(1)

END OF HOMEWORK **Total = 35 marks**
Hand your work in with this sheet. Failure to meet your target grade will result in intervention support.
File this question paper in your file, ready for revision and file inspection.

Answer booklet

3) *You may not need to use all the rows in this table*

x	y	t	Is x odd?	Is $x = 0$?
27	5	0	Yes	

4) *You may not need to use all the rows in this table*

N	**E**	**R**	**Is $-10^{-6} < R - E < 10^{-6}$?**